ARTI 502
Deep Learning: 2st Assignment
Training-Testing CNN with CIFAR Dataset using Pytorch
Year 2026 Term 1

Group #
Group Members:

#	Member Name	Member Academic ID
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Overview

In this assignment, we will have you train some convolutional neural networks and will use a small dataset (CIFAR).

Objective

To implement two core building blocks to deep learning:

- Convolution layers
- Maxpooling layers.

Tasks (with PyTorch)

Task 1: Getting the CIFAR Dataset

Instructions:

- Download the dataset first. Since PyTorch hosts the CIFAR dataset, you can load it using their helper function later.
- You need to change DATA_PATH to an empty directory to download the dataset to.

Deliverable:

Screenshot showing the dataset loaded successfully.

Task 2: Defining the Network

Instructions:

- Just like with MNIST in the first assignment you need to define your network architecture. This time you will be using convolutional layers and maxpooling to extract features from our images before you feed those features into your final classifier.

- Check out the documentation for nn.Conv2d

- So the first parameter is the number of channels in the input. Second is the number of filters we'll use (AKA number of channels in the output). Third is kernel size. Next is stride and padding which are optional and have default values.

- Create your training and test functions

Deliverable:

- Provide screenshots of your code to define your network
- Show screenshoots of your training and test functions.

Task 3: Loading Data

Instructions:

- It's worthwhile to check out how the dataloader loads the images into tensors. You can print out the size of the loaded data in the images tensors. Also you can use matplotlib to see the visualization of the images data

Deliverable:

- Provide screenshots how to find:
  1. Images tensor size
  2. Data images vizualization using matplotlib

Task 4: Dataset Augmentation

Instructions:

- In this assignment, you may find that data augmentation helps you a lot (or possibly hurts your performance).
- Play around with various data augmentations we will suggest some:

  1. ToPILImage - This one is useful for a lot of the built in transforms which expect PIL images.
  2. RandomHorizontalFlip
  3. RandomResizedCrop
  4. ColorJitter
  5. RandomRotation
  6. Normalize
  7. Adding various types of noise
  8. ToTensor - PyTorch expects the output from the dataset to be a tensor in CxHxW format.

  Note: You should be careful about which of these you apply to the test data. You usually don't want to apply noise to the test data, but you do want to normalize it the same way for example.

Deliverables:

- Apply your code with various Dataset Augmentation as mentioned above.

Task 5: Training Function and Tuning hyperparameters

Instructions:

- Generally, it is useful to see how your training is going. Often people print their loss to make sure it goes down and their accuracy to make sure it goes up. But pictures are better than words. So for this part, you should record and plot the training loss, test loss, and test accuracy (and whatever else you want).

- Try different network architectures and experiment with hyperparameters.

Deliverables:

- Screenshot of your python source code that contains your traing function with above requirements
- What design that you tried worked the best? This includes things like network design, learning rate, batch size, number of epochs, and other optimization parameters, data augmentation etc
- What was the final train loss? Test loss? Test Accuracy? Provide the plots for train loss, test loss, and test accuracy.

Task 6: Save your source code

Instructions:

- Save your python source code

Deliverable:

- Submit your Github link of your source code along with this document, which includes all screenshots via Blackboard.

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End of Assignment -

Grading Rubric

Criterion	Excellent	Good	Developing	Poor	Marks
Task 1: Getting the CIFAR Dataset	Dataset loaded correctly with clear screenshot showing successful execution	Loaded correctly but screenshot is unclear	Dataset loaded with issues or incomplete evidence	Not loaded / missing screenshot	**0.5**
Task 2: Defining the Network	All clear screenshots	Minor mistakes or missing small details	Several missing description or unclear screenshots	Not answered / incorrect / missing evidence	**2**
Task 3: Loading Data	All clear screenshots	Minor mistakes or missing small details	Several missing description or unclear screenshots	Not loaded / missing screenshot	**0.5**
Task 4: Dataset Augmentation	All datasets augmentation implemented	Major datasets augmentation implemented	Minor datasets augmentation implemented	Not done	**2**

Task 5: Training Function and Tuning parameters	Tuning parameters and model architecture created, screenshot included, and resultlts description explained accurately	Tuning Parameters and model architecture correct but results description is weak	Tuning Parameters and model architecture corect but incorrect results description	Not done	**2**
Task 6: Save your source code	Source code save succefully	Source code save succefully done but minor missing item	Source code save incomplete or file format incorrect	Not submitted / missing	**0.5**
Total					**7.5**